USER CENTERED DESIGN

UNIT-I: INTRODUCTION TO HUMAN COMPUTER INTERACTION

SYLLABUS CONTENTS:

Includes the difference between good and poor interaction design, what interaction design is and how it relates to human-computer interaction and other fields, what is involved in the process of interaction design, the different forms of guidance used in interaction design, etc. Need for Design - Examples from Design of everyday things, case studies, Evolution of the web and digital interfaces, Design thinking and wicked problems.

Exercise - Identify problems around us requiring design solutions Or problems solved using design.

INTRODUCTION

Human computer interaction (HCI), alternatively man machine interaction (MMI) or computer human interaction (CHI) is the study of interaction between people (users) and computers.

With today's technology and tools, and our motivation to create effective and usable interfaces and screens, why do we continue to produce systems that are inefficient and confusing or, at worst, just plain unusable? Is it because:

1. We don't care?
2. We don't possess common sense?
3. We don't have the time?
4. We still don't know what really makes good design?

DEFINITION

Human-computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them."

GOALS

- A basic goal of HCI is

_ to improve the interactions between users and computers

_ by making computers more usable and receptive to the user's needs.

- A long-term goal of HCI is

_ to design systems that minimize the barrier between the human's cognitive model of what they want

_ to accomplish and the computer's understanding of the user's task

WHY IS HCI IMPORTANT

- User-centered design is playing a crucial role!
- It is getting more important today to increase competitiveness via HCI studies (Norman, 1990)
- High-cost e-transformation investments
- Users lose time with badly designed products and services • Users even give up using bad interface

_ Ineffective allocation of resource

DEFINING THE USER INTERFACE

User interface, design is a subset of a field of study called human-computer interaction

(HCI).

Human-computer interaction is the study, planning, and design of how people and computers work together so that a person's needs are satisfied in the most effective way. HCI designers must consider a variety of factors:

what people want and expect, physical limitations and abilities people possess,

-how information processing systems work, _ what people find enjoyable and attractive.

_ Technical characteristics and limitations of the computer hardware and software must also be considered.

The *user interface* is to

_ the part of a computer and its software that people can see, hear, touch, talk to, or otherwise understand or direct.

The user interface has essentially two components: input and output.

Input is how a person communicates his / her needs to the computer.

_ Some common input components are the keyboard, mouse, trackball, one's finger, and one's voice.

Output is how the computer conveys the results of its computations and requirements to the user.

_ Today, the most common computer output mechanism is the display screen, followed by mechanisms that take advantage of a person's auditory capabilities: voice and sound. The use of the human senses of smell and touch output in interface design remains largely unexplored.

Proper interface design will provide a mix of well-designed input and output mechanisms that satisfy the user's needs, capabilities, and limitations in the most effective way possible.

The best interface is one that is not noticed, one that permits the user to focus on the information and task at hand, not the mechanisms used to present the information and perform the task.

GOOD DESIGN

Good UX design can benefit a business in numerous ways, including increased sales. Here are five examples:

- Professional, sleek designs can help a business stand out and attract consumers.
- A well-designed website can create a positive first impression to help establish lasting customer relationships.
- Easy website navigation can make the customer journey smoother and more enjoyable.
- As demonstrated by Google's recognizable text colors, consistent branding can benefit a business by increasing brand recognition.
- Good design can unlock the full potential of digital products and guide user actions.

THE IMPORTANCE OF GOOD DESIGN

- Good design is innovative.
- Good design makes a product useful.
- Good design is aesthetic.
- Good design makes a product understandable.
- Good design is unobtrusive.
- Good design is long-lasting.
- Good design is environmentally friendly

Redbus is website used for booking bus tickets online. It provides a quick and easy way to book bus tickets online. It is a good design as it is simple, easy to understand and use, unobtrusive and a useful product. Users can easily search for available bus services by entering the date of travel and place they want to travel to. They can then book the tickets online making it an easy and a seamless experience for booking bus tickets online.

THE IMPORTANCE OF THE USER INTERFACE

- A well-designed interface and screen are terribly important to our users. It is their window to view the capabilities of the system.
- It is also the vehicle through which many critical tasks are presented. These tasks often have a direct impact on an organization's relations with its customers, and its profitability.
- A screen's layout and appearance affect a person in a variety of ways. If they are confusing and inefficient, people will have greater difficulty in doing their jobs and will make more mistakes.

Poor design may even chase some people away from a system permanently. It can also lead to aggravation, frustration, and increased stress.

BAD DESIGN

Bad design refers to design that fails to meet user needs effectively, lacks functionality, or creates unnecessary complexity, which leads to frustration, inefficiency and a poor user experience. It often results from overlooking essential principles of good design, such as usability, accessibility, aesthetics, and user-friendliness.

The characteristics that typically contribute to a design being considered flawed include:

1. Lack of Clarity and Usability

Designs that confuse users, make navigation difficult, or obscure important information hinder usability. For instance, a website with a cluttered layout, confusing menus, or hidden call-to-action buttons can leave users feeling lost and frustrated.

2. Poor Accessibility

Design that does not consider all users, including those with disabilities, is a hallmark of bad design. This could be a website that fails to provide alternative text for images (crucial for screen readers), uses color combinations with low contrast, or lacks keyboard navigation, making it inaccessible to users with visual impairments or other disabilities.

3. Ignores User Needs and Feedback

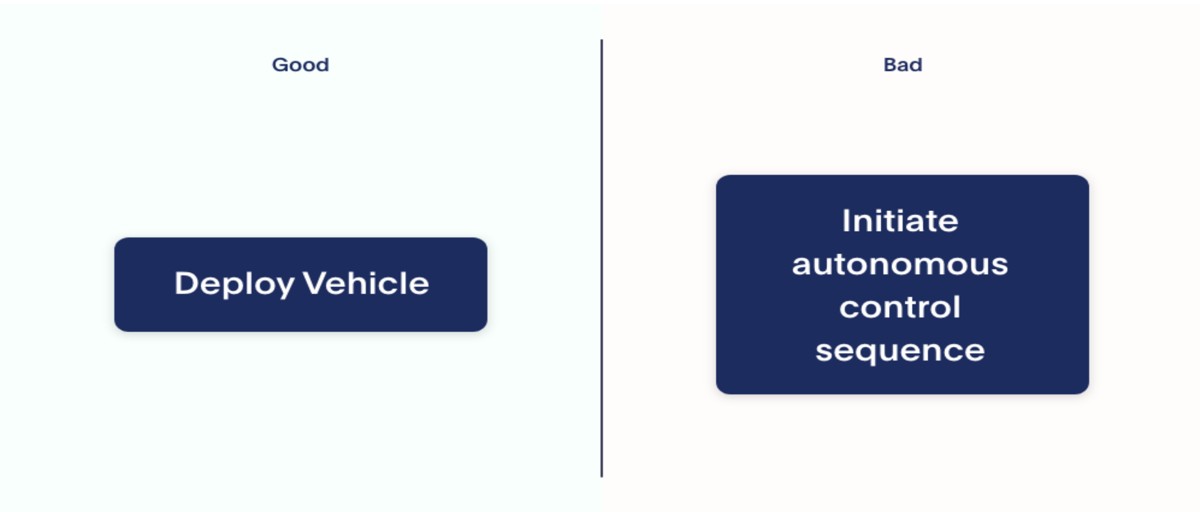
Designs created without understanding or considering the target audience's needs, preferences, and behaviours are likely to miss the mark.

The release of Windows 8 by Microsoft in 2012 is a notable example of a product criticized for ignoring user needs and feedback, primarily due to its drastic redesign that removed the familiar Start menu in favor of a new tile-based, touch-centric interface.

Bad Design example:

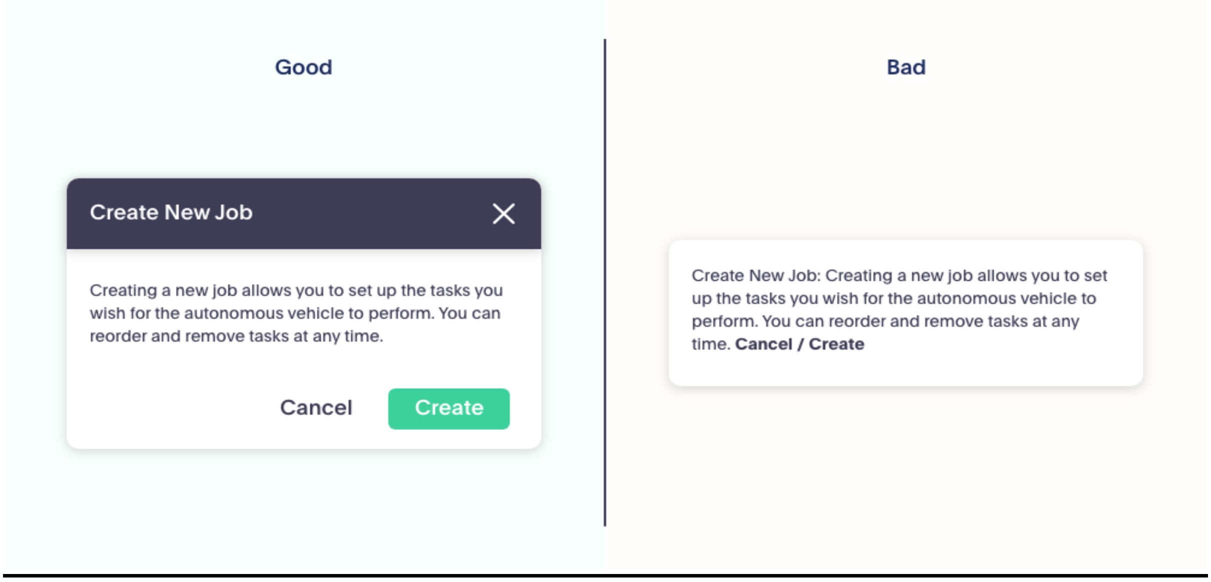
1. Button Information

It is essential to keep buttons concise. Large buttons will likely be interpreted as a text block or an ad rather than a call-to-action.



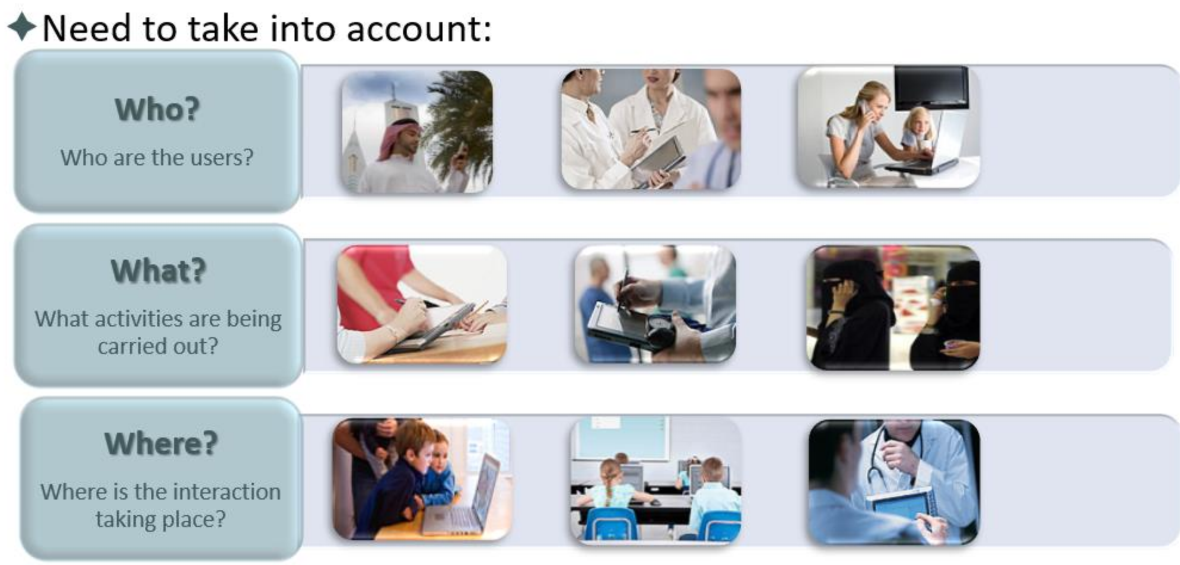
2. Pop-Up Boxes

These boxes are meant to explain concepts, call to action, or confirm a certain action. In the bad example, the box does not give the option to close the window, and the "Cancel/Create" buttons are too small to create a call to action. The good example has solved these two issues.



3. Real Time Example:

WHAT TO DESIGN



- Need to consider what people are good and bad at
- Consider what might help people in the way they currently do things.
- Think through what might provide quality user experiences.
- Listen to what people want and get them involved.
- Use tried and tested **user-centered methods.**

Interaction Design (IxD):

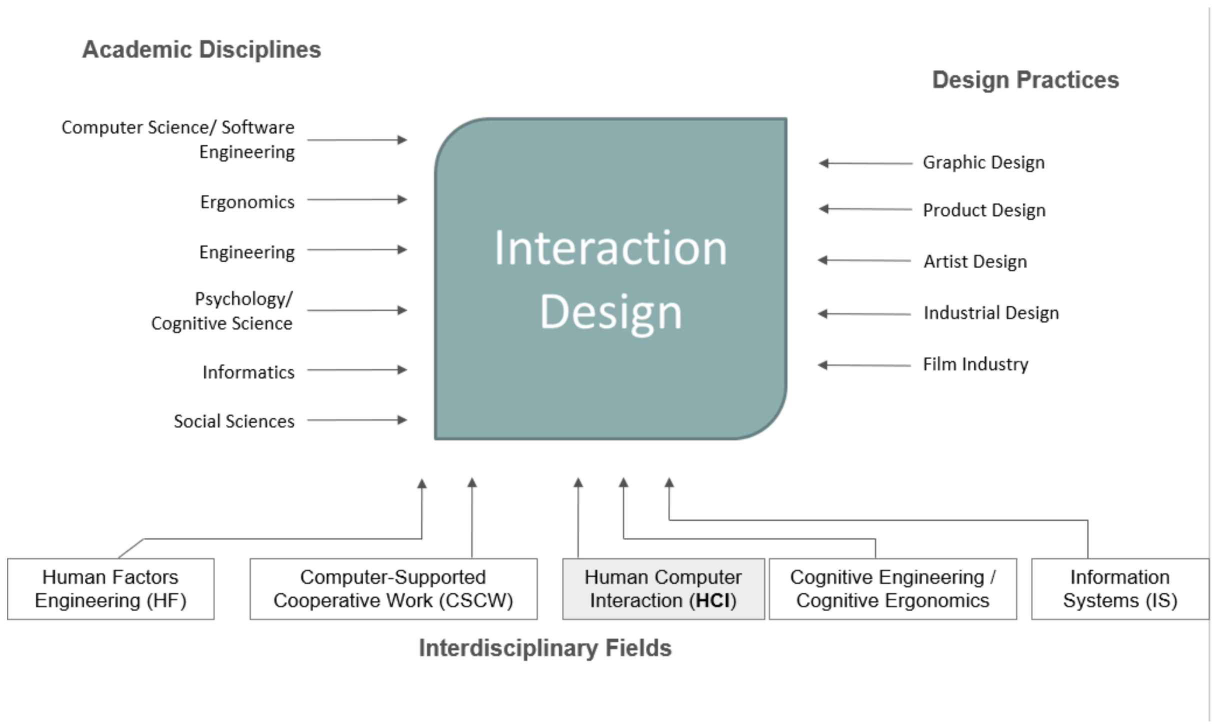
Interaction design is a multidisciplinary design field that focuses on the interaction between users and digital products, system or interfaces.

It involves designing how user engage with and experience a product. IxD focuses on HCI including animation, micro interactions, transitions, search and other motion based designs. They decide e.g. what happens when a user taps an element.

UI Design:

User interface design focuses on visual design and aesthetic, including colour, fonts, iconography, layouts etc. They decide what a user interface must look like.

HCI AND INTERACTION DESIGN:



Academic Disciplines:

- Computer science/ Software engineering: Computer science provides the technical foundations for developing interactive systems and applications.

- Ergonomics: The study of people in their working environment. The goal is to eliminate discomfort and risk of injury due to work.

- Psychology/ Cognitive Science: The influence of cognitive psychology on IxD is increasingly becoming a significant factor. Through the application of cognitive psychology, designers make informed decisions regarding what they should create in alignment with user concept, thoughts and other activities.

- Social Science: It offers understanding of cultural contexts, social behaviour and user communities, helping designers create interfaces.

- Informatics: This aspect of informatics deals with organizing and structuring information to enhance usability. Effective information architecture is crucial for IxD, as it helps designers create logical and intuitive pathways for users to navigate through interactive systems

Design Practices:

- Graphic Design: Graphic design is a vital discipline focused on visual communication and aesthetics. It involves creating visual content to convey messages effectively and is integral to various fields, including marketing, branding, and user interface design.

- Product Design: Product design is the process of creating new products for sale that meet the needs and desires of consumers. It encompasses the entire lifecycle of a product, from ideation and development to usability and market introduction.

- Artist Design: Artist design refers to the creative process involved in producing visual art, emphasizing the principles and techniques that artists use to convey their ideas and emotions. This discipline encompasses various forms of art, including illustration, concept art, and fine arts.

- Industrial Design: Focuses on physical form, ergonomics, material and manufacturing processes to create products that are functional, ergonomic and visually appealing.

- Film Industry: It involves designing interface for on-screen graphics, animations and visual effects that enhance storytelling and engage audience.

Interdisciplinary Fields:

- Human Factor Engineering: It focuses on understanding human capabilities, limitations and behaviour to design safe, efficient and effective system.

- Computer Supported Cooperative Work: Focuses on understanding how technology can support and enhance collaborative work and activities. Examines how people work together, share information and coordinate actions in various contexts.
  CSCW informs IxD by providing insights into collaborative work practices, social dynamics and communication patterns.

- Human-Computer Interaction: Informatics encompasses HCI as a key focus area. HCI studies the design and evaluation of user interfaces and interactions, which directly informs IxD practices. This relationship fosters a deeper understanding of how users interact with technology, guiding IxD in creating better designs
- Cognitive engineering /Cognitive Ergonomics: Cognitive engineering, often referred to as cognitive ergonomics, is a field that focuses on optimizing the interaction between humans and complex systems by understanding and supporting cognitive processes.

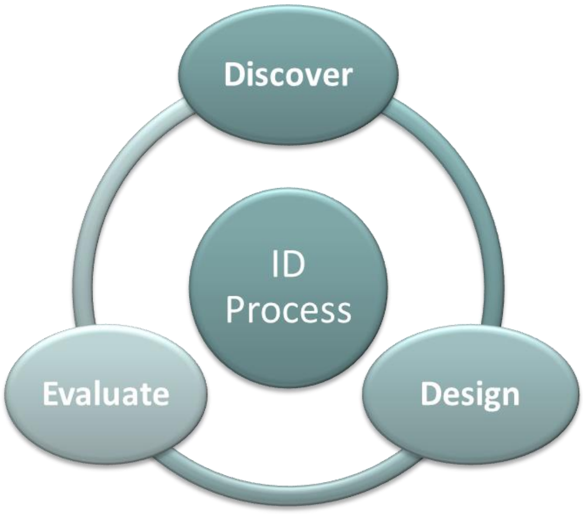
- Information Systems: An information system (IS) is a structured system designed to collect, store, manage, and disseminate data and information. It integrates technology, people, and processes to support decision-making, coordination, control, analysis, and visualization within an organization.

PROFESSIONALS IN ID BUSINESS:

- Interaction designers - people involved in the design of all the interactive aspects of a product
- Usability engineers - people who focus on evaluating products, using usability methods and principles
- Web designers - people who develop and create the visual design of websites, such as layouts
- Information architects - people who come up with ideas of how to plan and structure interactive products

- User experience designers (UX) - people who do all the above but who may also carry out field studies to inform the design of products

ID PROCESS:



Discover: Identifying needs and establishing requirements for the user experience

Design: Developing alternative designs to meet these requirements.Building interactive prototypes that can be communicated and assessed

Evaluate: Evaluating what is being built throughout the process and the user experience it offers.

CORE CHARACTERISTICS OF INTERACTION DESIGN

- Users should be involved through the development of the project.
- Specific usability and user experience goals need to be identified, clearly documented and agreed at the beginning of the project.
- Iteration is needed through the core activities.

DIFFERENT FORMS OF GUIDANCE USED IN INTERACTION DESIGN

1. **Affordances:** Properties of an object that suggest how it can be used.
   Example: A button that looks pressable or a handle that looks like it can be pulled.
2. **Signifiers:** Indicators that communicate where and how actions should take place.
   Example: Arrows, labels, or icons that indicate interactive elements.
3. **Feedback:** Information returned to the user about what action has been done and what has been accomplished.
   Example: Visual changes when a button is clicked, sounds, or haptic feedback.
4. **Constraints**: Limitations that prevent users from making errors.
   Example: Graying out inactive options or restricting input formats in forms.
5. **Mapping**: Definition: Relationship between controls and their effects in the world. The layout of stove controls that match the arrangement of burners.
6. **Consistency:** Uniformity of design elements across a system.
   Example: Using the same color for similar actions or maintaining the same layout structure across different screens.
7. **Documentation:** Written or visual information that guides users on how to use the system.
   Example: Help manuals, tooltips, or onboarding tutorials.
8. **User Flow:** The path taken by a user to complete a task within an application or website.
   Example: Step-by-step process for creating an account or checking out in an online store.
9. **Direct Manipulation:** Interaction style where users can directly manipulate objects on the screen.
   Example: Drag-and-drop functionality or pinch-to-zoom on touchscreens.
   Different forms of guidance used in interaction design.
10. **Affirmation and Confirmation:** Requiring users to confirm their actions to prevent errors.
    Example: "Are you sure you want to delete this file?" prompts or confirmation dialogs.
11. **Progressive Disclosure:** Presenting only the necessary information at any given time.

Example: Advanced settings hidden under a "More Options" button.

12. **Onboarding:** Introducing new users to the system's features and functionalities.

    Example: Guided tours or interactive tutorials when a user first opens an app.

13. **Visual Hierarchy:** Organizing content to guide users' attention to the most important elements first.

    Example: Larger, bolder headings for primary information and smaller text for secondary details.

14. **Metaphors:** Using familiar concepts to help users understand new functions.

    Example: A trash can icon for deleting files or a shopping cart for purchasing items online.

Need for design: Examples from design of everyday things

- The need for design is fundamental in creating products, systems, and experiences that are functional, efficient, and satisfying for users. Here are some key reasons why design is essential:

1.Usability:

- Design ensures that products and interfaces are easy to use and understand.
- Good design reduces the learning curve and minimizes errors, enhancing overall user satisfaction.

2. Efficiency:

- Well-designed products and systems enable users to achieve their goals with minimal effort and time.
- Efficiency in design can increase productivity and reduce frustration, leading to a more positive user experience

3. Aesthetics:

- Aesthetically pleasing designs create a positive emotional response and enhance user engagement.
- Attractive design can increase the perceived value of a product and make it more appealing to users.

4. Accessibility:

- Design ensures that products are usable by people with a wide range of abilities and disabilities.
- Inclusive design broadens the user base and ensures that everyone can benefit from the product or service.

5.Safety:

- Good design minimizes the risk of accidents and injuries.
- Designing for safety is crucial in environments where users interact with potentially hazardous products or systems.

6.User Satisfaction:

- Design contributes to the overall satisfaction of users by meeting their needs and expectations.
- Satisfied users are more likely to remain loyal to a product or brand and recommend it to others.

7. Innovation:

- Design drives innovation by exploring new ideas and approaches to problem-solving.
- Innovative design can lead to the development of groundbreaking products and services that meet emerging needs and create new markets.

8.Brand Identity:

- Consistent and thoughtful design helps build a strong brand identity.
- A distinctive design can differentiate a brand from its competitors and foster brand loyalty.

9. Market Success:

- Products that are well-designed are more likely to succeed in the market.
- Good design can be a significant competitive advantage, leading to higher sales and better market performance

10.Sustainability:

- Design can contribute to environmental sustainability by promoting efficient use of resources and reducing waste.
- Sustainable design practices are essential for addressing environmental challenges and creating products that are environmentally friendly.

Evolution of the web and digital interfaces:

The evolution of the web and digital interfaces has been marked by significant technological advancements and changes in user expectations. Here's a comprehensive overview of this evolution:

1. Early Web (1990s)

- **Web 1.0**:
  - **Characteristics**: Static pages, text-heavy, limited interactivity.
  - **Technologies**: HTML, HTTP, simple browsers (Netscape Navigator, early versions of Internet Explorer).
  - **Design**: Basic, text-oriented with hyperlinks for navigation. Images were minimal due to bandwidth limitations.

2. Emergence of Dynamic Content (Late 1990s to Early 2000s)

- **Web 1.5**:
  - **Characteristics**: Introduction of server-side scripting and dynamic content generation.
  - **Technologies**: CGI, Perl, PHP, ASP.
  - **Design**: Still largely text-based, but with more interactive elements like guest books and basic forms.

3. Web 2.0 (Mid-2000s)

- **Characteristics**: User-generated content, increased interactivity, social networking.
- **Technologies**: AJAX, JavaScript, CSS, XML, RSS feeds.
- **Design**: More visually appealing with CSS for styling, user-centric design, emergence of blogs, wikis, and social media platforms (MySpace, Facebook, YouTube).

4. Rich Internet Applications (Late 2000s)

- **Characteristics**: Enhanced user experiences with rich media content.
- **Technologies**: Adobe Flash, Silverlight, advanced JavaScript frameworks (jQuery).
- **Design**: Multimedia elements, animations, and interactive components became more common.

5. Mobile Revolution (Late 2000s to Early 2010s)

- **Characteristics**: Growth of mobile web access and responsive design.
- **Technologies**: HTML5, CSS3, responsive design frameworks (Bootstrap), mobile operating systems (iOS, Android).
- **Design**: Emphasis on mobile-first design, touch-friendly interfaces, and adaptive layouts to cater to different screen sizes.

6. Web 3.0 and Semantic Web (2010s)

- **Characteristics**: Focus on data interoperability, machine-readable content, and personalized user experiences.
- **Technologies**: Semantic HTML, RDF, SPARQL, microdata.
- **Design**: Cleaner, more semantic code structure, enhanced accessibility features, and personalized content delivery.

7. Modern Web and Progressive Web Apps (2015 onwards)

- **Characteristics**: Blurring lines between web and native apps, offline capabilities, and push notifications.
- **Technologies**: Service Workers, WebAssembly, PWA frameworks (Angular, React, Vue.js).
- **Design**: App-like experiences on the web, minimalistic and performance-focused designs, use of Material Design and other design systems.

8. Rise of AI and Voice Interfaces (Late 2010s to Present)

- **Characteristics**: Integration of artificial intelligence, voice recognition, and virtual assistants.
- **Technologies**: AI APIs (Google Assistant, Alexa), natural language processing, machine learning.
- **Design**: Voice user interfaces (VUIs), conversational design, and smart assistants becoming part of the user experience.

9. Future Trends

- **Characteristics**: Continued integration of AI, augmented reality (AR), virtual reality (VR), and the Internet of Things (IoT).
- **Technologies**: WebXR, 5G, blockchain, decentralized web technologies.
- **Design**: Immersive experiences, real-time interactivity, more personalized and context-aware interfaces, and enhanced security and privacy measures.

Key Concepts in the Evolution of Web and Digital Interfaces:

1. **User-Centered Design**: Focus on understanding and meeting user needs.
2. **Accessibility**: Ensuring digital interfaces are usable by people with diverse abilities.
3. **Responsive Design**: Creating interfaces that adapt to various screen sizes and devices.
4. **Performance Optimization**: Enhancing speed and efficiency of web interactions.
5. **Security**: Implementing robust security measures to protect user data and privacy.

The web and digital interfaces have come a long way from static pages to dynamic, interactive, and intelligent experiences. As technology continues to advance, we can expect even more innovative and user-centric developments in the future.

Design thinking and wicked problems:

Wicked problems:

Wicked problems are complex, multifaceted issues that are difficult to define and inherently resistant to resolution. The term was first coined by Horst Rittel and Melvin Webber in the 1970s, and it refers to problems that involve numerous interdependent factors, making them seem almost impossible to solve.

Characteristics of Wicked Problems

No Definitive Solution: There is no clear formula or definitive solution to wicked problems. Each problem is unique and can be approached in various ways.

Complexity and Interconnectedness: Wicked problems are often interconnected with other issues, complicating efforts to isolate and solve them.

Diverse Stakeholder Perspectives: Different stakeholders may have conflicting interests and interpretations of the problem, leading to varied explanations and potential solutions.

Evolving Nature: Wicked problems can change over time, requiring adaptive solutions that can evolve with the problem.

One-Shot Operations: Solutions to wicked problems are often irreversible and cannot be tested in a straightforward manner, meaning that every attempt to solve them is significant.

No Stopping Rule: There is no clear endpoint to the problem-solving process, making it difficult to determine when a solution is "final" or "successful".

Design Thinking as a Solution Approach

Design thinking is a human-centered approach to innovation that can be particularly effective in tackling wicked problems. It involves understanding user needs, reframing problems, brainstorming solutions, prototyping, and testing. Here’s how design thinking can be applied to wicked problems:

1. Empathy and Understanding

Engaging with stakeholders to understand their perspectives and needs is crucial. This involves active listening and empathy to grasp the complexities of the problem.

2. Defining the Problem

Rather than jumping to solutions, design thinking encourages a deep exploration of the problem space. This may involve reframing the problem to uncover underlying issues.

3. Ideation and Brainstorming

Generating a wide range of ideas allows for creative solutions to emerge. This phase encourages collaboration and the inclusion of diverse viewpoints.

4. Prototyping

Developing low-fidelity prototypes helps visualize solutions and allows for experimentation. Prototyping can reveal unforeseen challenges and opportunities.

5. Testing and Iteration

Solutions are tested in real-world contexts, and feedback is gathered to refine and improve the approach continuously. This iterative process is essential for adapting to changing conditions and stakeholder needs.

Real-World Examples

Case Study: Urban Transportation Challenges

Problem Identification

Urban areas worldwide face significant transportation challenges, including congestion, pollution, and accessibility. These issues impact daily commutes, public health, and the overall quality of life for residents.

Design Solutions Implemented

1. Bicycle Sharing Systems

Example: Citi Bike in New York City

Problem: High traffic congestion and pollution from vehicles.

Design Solution: The implementation of a bike-sharing program that provides residents and tourists with access to bicycles for short trips.

Impact:

Reduced traffic congestion by encouraging cycling as a viable transportation option.

Improved air quality by decreasing reliance on cars.

Enhanced accessibility to public transport hubs, making it easier for people to reach their destinations.

Example 1: Climate Change

Problem: Climate change is a classic wicked problem involving environmental, economic, and social factors.

Design Thinking Approach: Initiatives like the Carbon Trust utilize design thinking to engage stakeholders, develop innovative solutions for carbon reduction, and promote sustainable practices across industries.

Example 2: Urban Transportation

Problem: Urban transportation systems often face congestion, pollution, and inequity.

Design Thinking Approach: Cities like Barcelona have adopted design thinking to create smart traffic management systems that adapt to real-time conditions, improving traffic flow and reducing emissions while considering the needs of various stakeholders.

Example 3: Healthcare

Problem: Access to healthcare is a wicked problem influenced by socioeconomic factors, policy, and individual needs.

Design Thinking Approach: Organizations like IDEO have applied design thinking to develop patient-centered healthcare solutions, improving access and quality of care through innovative service models.

CASE STUDY: URBAN TRANSPORTATION CHALLENGES

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2. Smart Traffic Management Systems

Example: Barcelona's Smart Traffic Lights

Problem: Inefficient traffic flow leading to congestion and delays.

Design Solution: Installation of smart traffic lights that adapt to real-time traffic conditions using sensors and data analytics.

Impact:

Improved traffic flow by reducing wait times at intersections.

Decreased emissions from idling vehicles, contributing to better air quality.

Enhanced safety for pedestrians and cyclists through better traffic management.

3. Public Transportation Apps

Example: Transit App

Problem: Difficulty in navigating public transportation systems, leading to underutilization.

Design Solution: Development of a mobile application that provides real-time information about public transit schedules, routes, and delays.

Impact:

Increased ridership on public transport due to improved accessibility and user experience.

Empowered users with information, making it easier to plan their journeys.

Reduced reliance on personal vehicles, contributing to lower traffic congestion and pollution.

Design Process and Techniques Used

User-Centered Design: Each solution was developed with a focus on the end user, ensuring that the designs met the actual needs and preferences of the community.

Prototyping and Testing: For instance, before launching Citi Bike, pilot programs were conducted to gather user feedback and refine the system.

Data-Driven Decision Making: Smart traffic systems utilize data analytics to continuously improve performance based on real-time traffic conditions.

Sustainability Considerations: Each design solution aimed to promote sustainable practices, such as reducing carbon footprints and encouraging eco-friendly transportation modes.

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